Activity 3.2 – RISC-V Program Examples

1.Design Approach:

Designed Verilog modules demonstrating arithmetic algorithms such as summation, maximum value selection, greatest common divisor (GCD) and multiplication. Implemented combinational logic to calculate the required outputs.

2.Verification Steps:

- Created a separate testbench.
- Applied different input operands.
- Tested summation calculation.
- Verified maximum value selection.
- Tested GCD calculation.
- Checked multiplication results.
- Simulated using ModelSim.

3.Results:

The testbench checks the outputs of the implemented arithmetic algorithms. PASS/FAIL messages are displayed.

4.Issues Faced:

Arithmetic expressions must use appropriate operand widths. GCD calculation requires correct termination conditions. Multiplication results must be checked for overflow.

5.Conclusion:

Successfully designed and verified arithmetic algorithm demonstration modules using Verilog HDL.